

SQL Trigger Examples for Students and Departments Table

Table Definitions (For Reference)

```
CREATE TABLE Department (  
    dept_id INT PRIMARY KEY,  
    dept_name VARCHAR(50)  
);  
  
CREATE TABLE Student (  
    student_id INT AUTO_INCREMENT PRIMARY KEY,  
    name VARCHAR(50),  
    dept_id INT,  
    FOREIGN KEY (dept_id) REFERENCES Department(dept_id)  
);
```

1. BEFORE INSERT Trigger Prevent inserting a student with a NULL name

```
DELIMITER //  
  
CREATE TRIGGER before_student_insert  
BEFORE INSERT ON Student  
FOR EACH ROW  
BEGIN  
    IF NEW.name IS NULL THEN  
        SIGNAL SQLSTATE '45000'  
        SET MESSAGE_TEXT = 'Student name cannot be NULL';  
    END IF;  
END;  
//  
DELIMITER ;
```

2. AFTER INSERT Trigger Log student insert into a log table

```
-- Create log table  
CREATE TABLE Student_Log (  
    log_id INT AUTO_INCREMENT PRIMARY KEY,  
    student_name VARCHAR(50),  
    action_time TIMESTAMP DEFAULT CURRENT_TIMESTAMP  
);  
  
-- Trigger  
DELIMITER //  
CREATE TRIGGER after_student_insert  
AFTER INSERT ON Student  
FOR EACH ROW  
BEGIN  
    INSERT INTO Student_Log(student_name)  
    VALUES (NEW.name);  
END;  
//
```

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```
DELIMITER ;
```

3. BEFORE UPDATE Trigger Prevent changing department

```
DELIMITER //
```

```
CREATE TRIGGER before_student_update
BEFORE UPDATE ON Student
FOR EACH ROW
BEGIN
    IF OLD.dept_id != NEW.dept_id THEN
        SIGNAL SQLSTATE '45000'
        SET MESSAGE_TEXT = 'Changing department is not allowed';
    END IF;
END;
//
DELIMITER ;
```

4. AFTER DELETE Trigger Log deleted students

```
-- Create deleted students log
CREATE TABLE Deleted_Students (
    student_id INT,
    name VARCHAR(50),
    deleted_on TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);

-- Trigger
DELIMITER //
CREATE TRIGGER after_student_delete
AFTER DELETE ON Student
FOR EACH ROW
BEGIN
    INSERT INTO Deleted_Students(student_id, name)
    VALUES (OLD.student_id, OLD.name);
END;
//
DELIMITER ;
```